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EE1030/EE2100: Matrix Theory

Lecture 21: Adjoint and Linear Transformation

Topics :

1. Adjoint of a Matrix
 2. Linear Transformation
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Hadamard Product of Two Matrices

Def: Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{m \times n}$. The Hadamard product of A & B , denoted by

$$(A \odot B)_{ij} = a_{ij} b_{ij}$$

$$A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \quad B = \begin{bmatrix} b_{11} & \dots & b_{1n} \\ \vdots & & \vdots \\ b_{m1} & \dots & b_{mn} \end{bmatrix}$$

$$A \odot B = \begin{bmatrix} a_{11} b_{11} & \dots & a_{1n} b_{1n} \\ \vdots & & \vdots \\ \underline{a_{m1} b_{m1}} & \dots & a_{mn} b_{mn} \end{bmatrix}$$

Applications :

- a) Applying color filters to images
- b) Control: Participation factors (Hadamard product of matrices whose columns/rows are eigen vectors)

Outer Product of Two Vectors

Def:- Let $\vec{v} \in \mathbb{R}^n$ & $\vec{u} \in \mathbb{R}^n$. The product $\vec{u} \vec{v}^T \in \mathbb{R}^{n \times n}$ & the entries are given by

$$\begin{matrix} \vec{u} \vec{v}^T \in \mathbb{R}^{n \times n} \\ \downarrow \\ \mathbb{R}^{n \times 1} \mathbb{R}^{1 \times n} \end{matrix}$$

$$(\vec{u} \vec{v}^T)_{ij} = u_i v_j$$

$$\begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix} = \begin{bmatrix} u_1 v_1 & u_1 v_2 & \dots & u_1 v_n \\ \vdots & \vdots & & \vdots \\ u_n v_1 & u_n v_2 & \dots & u_n v_n \end{bmatrix} = \begin{bmatrix} \vec{r}_1^T \\ \vdots \\ \vec{r}_n^T \end{bmatrix} = [\vec{r}_1 \dots \vec{r}_n]$$

$$\text{Trace}(\vec{u} \vec{v}^T) = \vec{u}^T \vec{v}$$

The rows of $\vec{u} \vec{v}^T$ are scaled versions of $\vec{v}^T$ where the scaling factor of row i is u_i .

The columns of $\vec{u} \vec{v}^T$ are scaled versions of $\vec{u}$ where the scaling factor of column i is v_i .

are $(\vec{r}_1, \dots, \vec{r}_n)$ linearly independent? $\rightarrow$ No,

No. of linearly independent vectors = 1.

$\vec{u} \vec{v}^T \in \mathbb{R}^{n \times n}$ matrix & $\text{Rank}(\vec{u} \vec{v}^T) = 1$.

Adjoint of a Matrix

Definition: Let $A \in \mathbb{R}^{m \times n}$, $\vec{x} \in \mathbb{R}^n$, $\vec{y} \in \mathbb{R}^m$. The Adjoint of A , A^* is defined as

$$\langle A\vec{x}, \vec{y} \rangle = \langle \vec{x}, A^* \vec{y} \rangle$$

Leading to the def:

$$A\vec{x} \in \mathbb{R}^m$$

$$\text{Let } B \in \mathbb{R}^{n \times m}$$

$$\langle A\vec{x}, \vec{y} \rangle \in \mathbb{R}$$

$$B\vec{y} \in \mathbb{R}^n$$

$$\langle \vec{x}, B\vec{y} \rangle \in \mathbb{R}$$

B is adjoint of A if

$$\textcircled{1} \quad \leftarrow \langle A\vec{x}, \vec{y} \rangle = \langle \vec{x}, B\vec{y} \rangle \quad \forall \vec{x} \in \mathbb{R}^n \text{ \& } \vec{y} \in \mathbb{R}^m$$

$$\text{LHS of } \textcircled{1} : A\vec{x} = \sum_{i=1}^n x_i \vec{a}_i \quad \text{where } \vec{a}_i \in \mathbb{R}^m \text{ (column vector of } A)$$

$$\langle A\vec{x}, \vec{y} \rangle = \sum_{j=1}^m y_j \left(\sum_{i=1}^n x_i \vec{a}_i \right)_j = \sum_{j=1}^m \sum_{i=1}^n y_j x_i a_{ji} \rightarrow \textcircled{3}$$

$x_i a_{ji}$

$$\text{RHS of } \textcircled{1} \quad B\vec{y} = \sum_{j=1}^m y_j \vec{b}_j \quad \text{where } \vec{b}_j \in \mathbb{R}^n \text{ (column vector of } B)$$

$$\langle \vec{x}, B\vec{y} \rangle = \sum_{i=1}^n x_i \left(\sum_{j=1}^m y_j \vec{b}_j \right)_i = \sum_{i=1}^n \sum_{j=1}^m x_i y_j b_{ij} \rightarrow \textcircled{4}$$

for $\textcircled{3} = \textcircled{4}$

Trivial solution $\rightarrow b_{ij} = a_{ji} \Rightarrow B = A^T$

$\hookrightarrow$ is this the only solution?

Adjoint of a Matrix

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Proof of uniqueness of A^* : $\vec{y} = \vec{e}_j \in \mathbb{R}^m$ and $\vec{x} = \vec{f}_i \in \mathbb{R}^n = \begin{bmatrix} 0 \\ \vdots \\ 1 \end{bmatrix}$ i^{th} entry.

$$A\vec{x} = \vec{a}_i$$

$$\langle A\vec{x}, \vec{y} \rangle = a_{ji} \rightarrow$$

$$A^* = B \mid B\vec{y} = \vec{b}_j$$

$$b_{ji} = a_{ji}$$

$$\langle B\vec{y}, \vec{x} \rangle = b_{ji} \rightarrow$$

Linear Transformation - Introduction

Matrix Vector Product : $\overset{\rightarrow \in \mathbb{R}^{m \times n}}{A} \vec{x} \rightarrow$ View this as an operation on $\vec{x}$

$$\vec{f}(\vec{x}) = A\vec{x} : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

Trivial to prove: $\vec{f}(\vec{x})$ is linear since

$$\begin{aligned} \text{a) } \vec{f}(\vec{x} + \vec{y}) &= A(\vec{x} + \vec{y}) = A\vec{x} + A\vec{y} \\ &= \vec{f}(\vec{x}) + \vec{f}(\vec{y}) \end{aligned}$$

$$\begin{aligned} \text{b) } \vec{f}(\alpha \vec{x}) &= A(\alpha \vec{x}) = \alpha A\vec{x} \\ &= \alpha \vec{f}(\vec{x}) \end{aligned}$$

$\therefore \vec{f}(\vec{x}) = A\vec{x}$ is linear.